

# Spectral Analysis Report

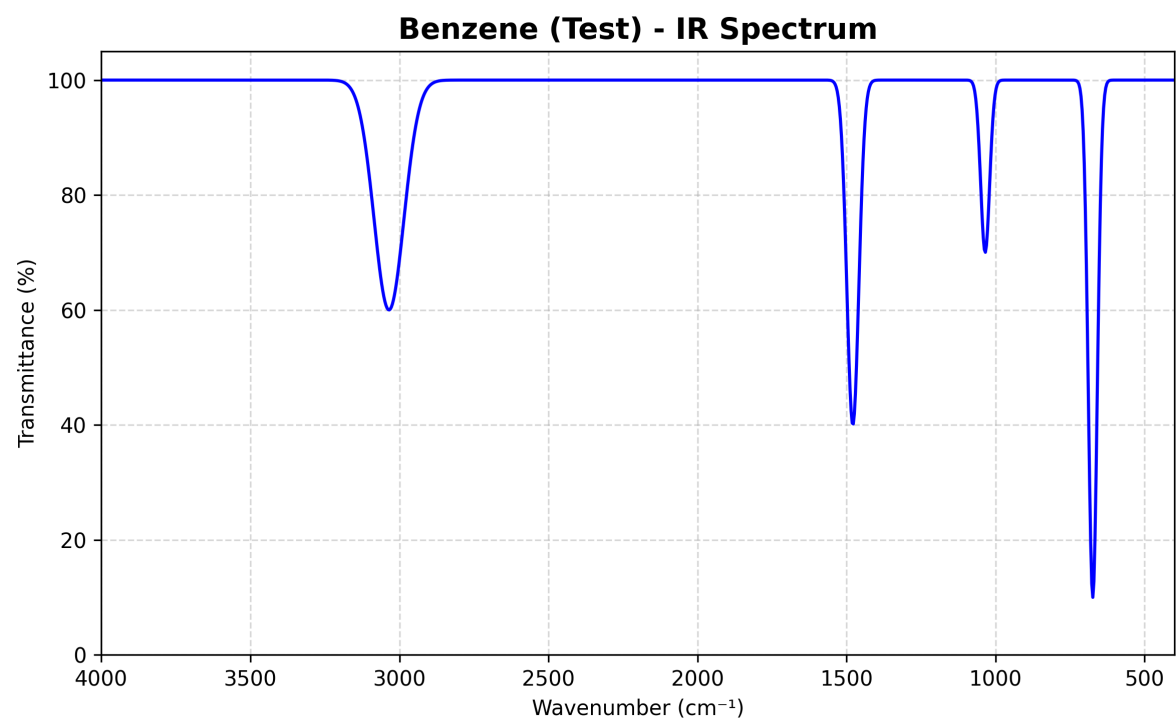
**Molecule: Benzene (Test)**

Date: 2026-03-02 05:08:59

## Molecular Structure

[No structure image provided]

# Infrared Spectrum (IR)

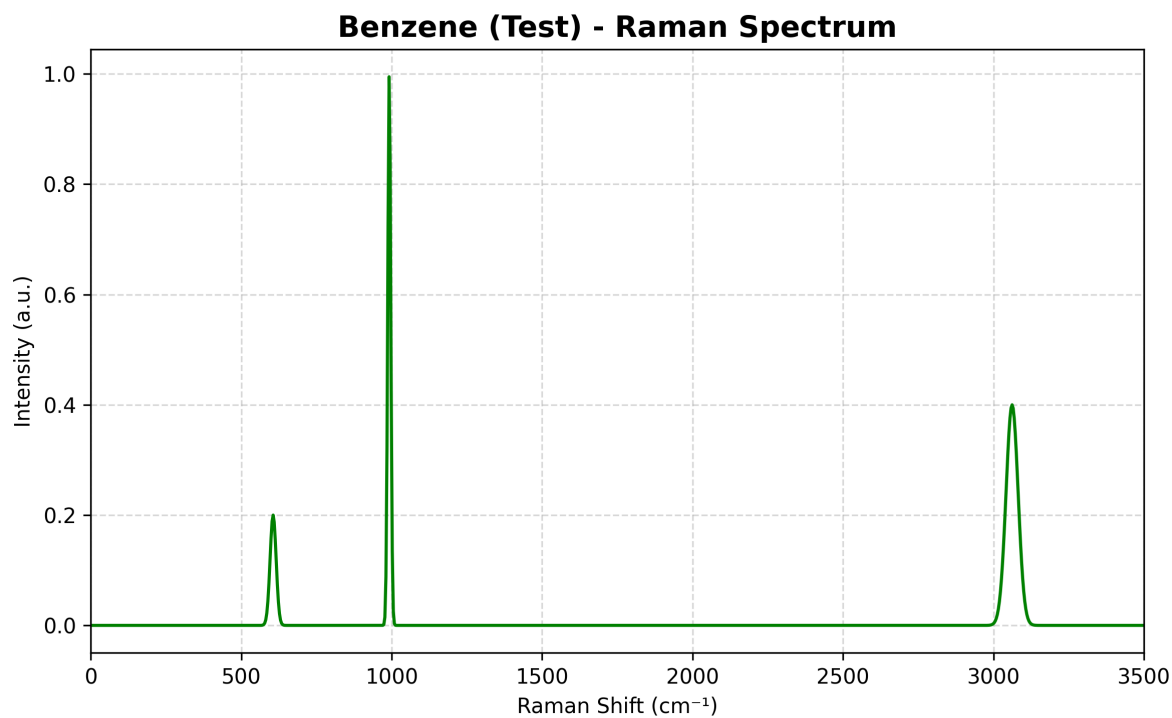


**Peak List:**

| Frequency / Shift | Intensity |
|-------------------|-----------|
| 3035.00           | 60.00     |
| 1479.00           | 40.00     |
| 674.00            | 10.00     |

Interpretation: Corrected: No C=O peak. Major peaks at 3035 (C-H) and 674 (C-H out-of-plane).

# Raman Spectrum



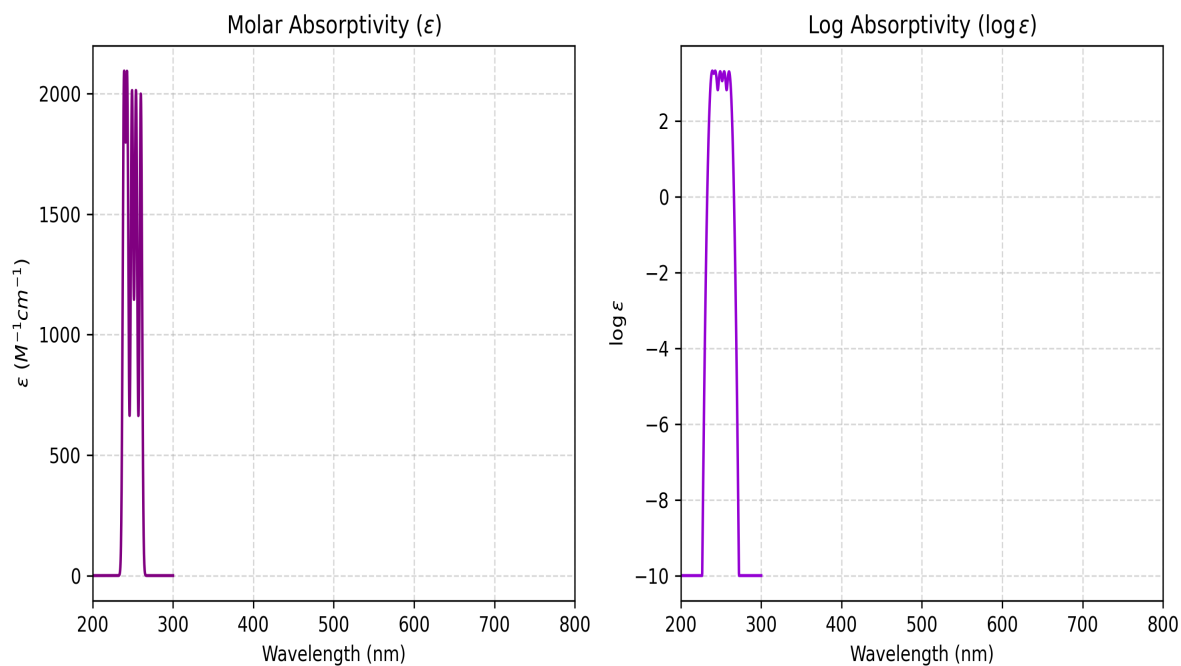
**Peak List:**

| Frequency / Shift | Intensity |
|-------------------|-----------|
| 3062.00           | 0.40      |
| 992.00            | 1.00      |

Interpretation: Dominant ring breathing mode at 992 cm-1 (VS).

# UV-Vis Spectrum

## Benzene (Test) - UV-Vis Spectrum

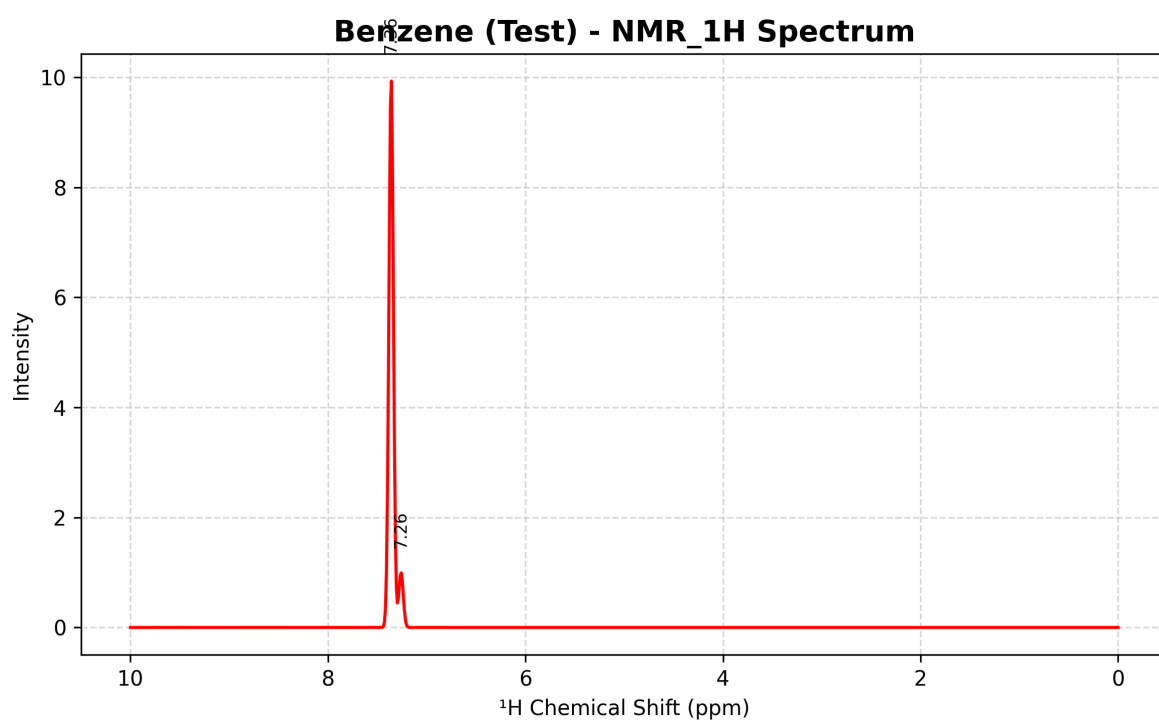


**Peak List:**

| Frequency / Shift | Intensity |
|-------------------|-----------|
| 254.00            | 0.20      |

Interpretation: Characteristic fine structure (B-band) centered at 254 nm.

# $^1\text{H}$ NMR Spectrum

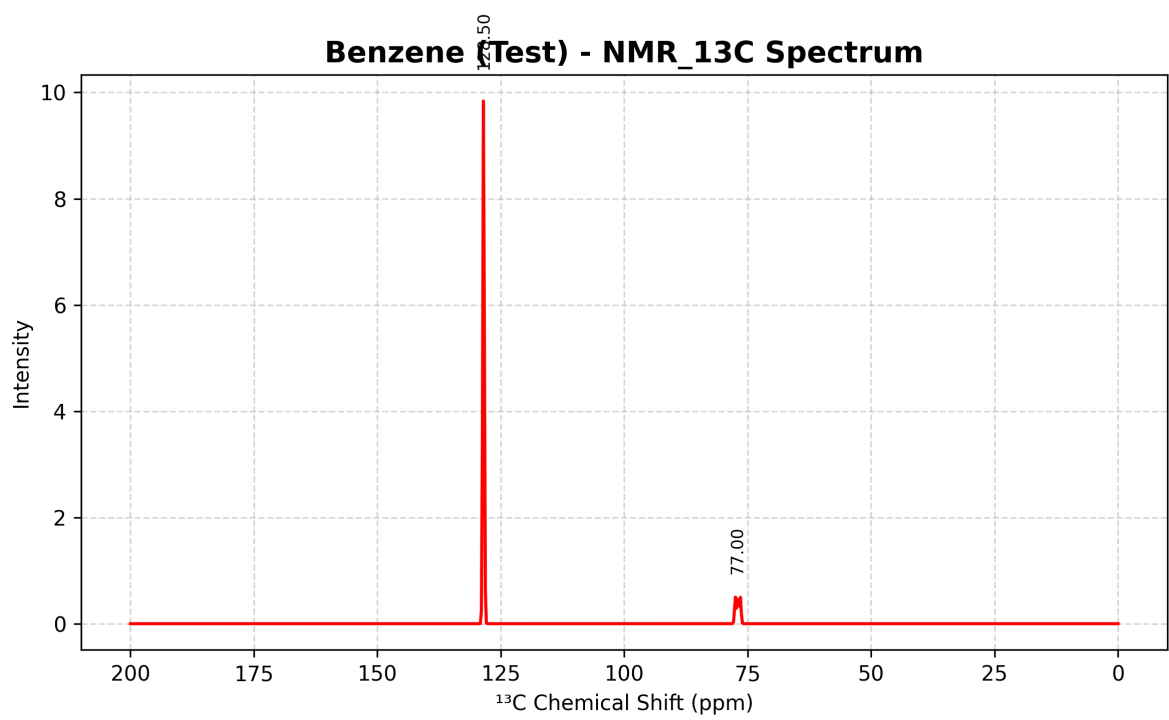


## Peak List:

| Frequency / Shift | Intensity |
|-------------------|-----------|
| 7.36              | 10.00     |
| 7.26              | 1.00      |

Interpretation: Benzene singlet at 7.36 ppm. Residual  $\text{CDCl}_3$  at 7.26 ppm.

<sup>13</sup>C NMR Spectrum



**Peak List:**

| Frequency / Shift | Intensity |
|-------------------|-----------|
| 128.50            | 10.00     |
| 77.00             | 0.50      |

Interpretation: Benzene carbon at 128.5 ppm. Solvent triplet at 77 ppm.